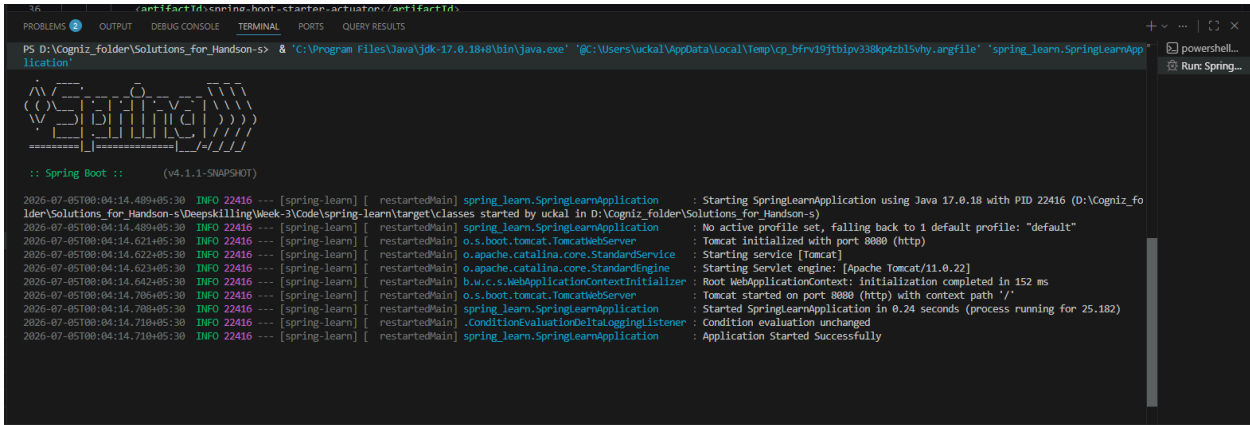


Spring REST using Spring Boot 3

1. Spring-rest-handson

Create a Spring Web Project using Maven

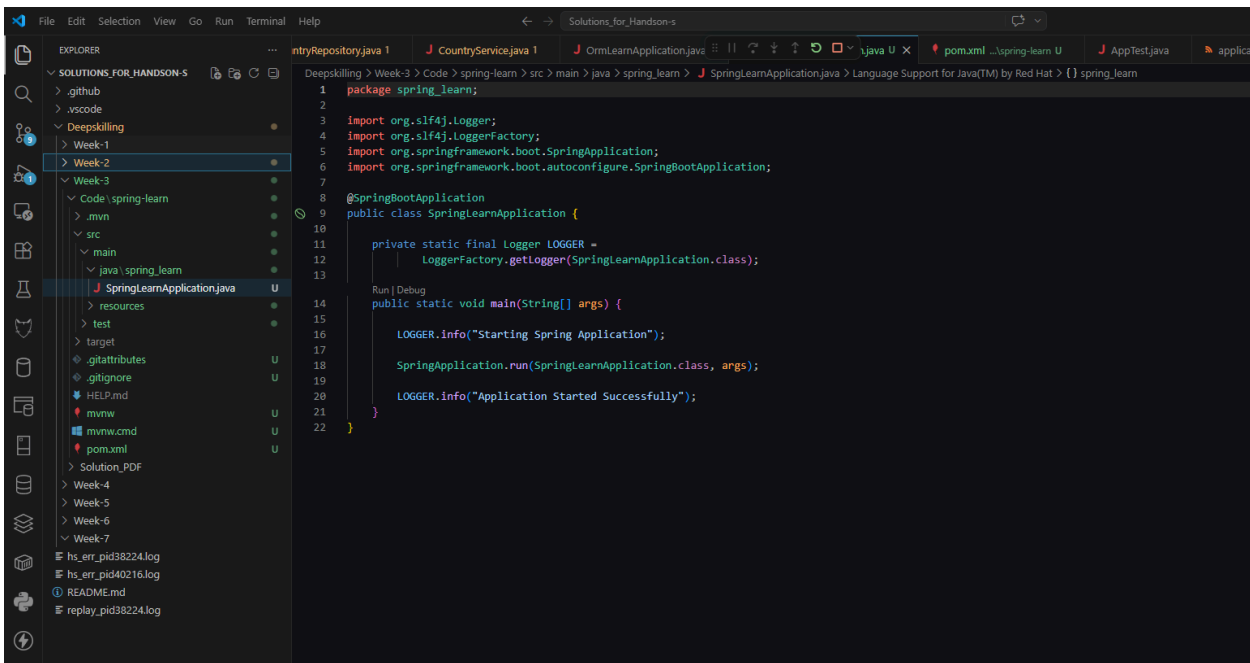


```
PS D:\Cogniz_folder\Solutions_for_Handson> & "C:\Program Files\Java\jdk-17.0.18\bin\java.exe" "@C:\Users\uckal\AppData\Local\Temp\cp_bfrv19jtb1pv338kp4z515why_argfile" 'spring_learn.SpringLearnApp
lication'

  ____  __  __
 / ___/  / /  / 
/ /   / /_/ / 
/ ___/  / ___/ 
/ /   / /   / 
/ /___/ /___/ 
\___/____/___/ 

:: Spring Boot ::
(v4.1.1-SNAPSHOT)

2026-07-05T00:04:14.489+05:30 INFO 22416 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.18 with PID 22416 (D:\Cogniz_fo
lder\Solutions_for_Handson-s\Deepskilling\Week-3\Code\spring-learn\target\classes started by uckal in D:\Cogniz_folder\Solutions_for_Handson-s)
2026-07-05T00:04:14.489+05:30 INFO 22416 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T00:04:14.624+05:30 INFO 22416 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-05T00:04:14.624+05:30 INFO 22416 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T00:04:14.624+05:30 INFO 22416 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T00:04:14.642+05:30 INFO 22416 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 152 ms
2026-07-05T00:04:14.706+05:30 INFO 22416 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-05T00:04:14.710+05:30 INFO 22416 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.24 seconds (process running for 25.182)
2026-07-05T00:04:14.710+05:30 INFO 22416 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition evaluation unchanged
2026-07-05T00:04:14.710+05:30 INFO 22416 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Application Started Successfully
```



```
1 package spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7
8 @SpringBootApplication
9 public class SpringLearnApplication {
10
11     private static final Logger LOGGGER =
12         LoggerFactory.getLogger(SpringLearnApplication.class);
13
14     Run | Debug
15     public static void main(String[] args) {
16
17         LOGGGER.info("Starting Spring Application");
18
19         SpringApplication.run(SpringLearnApplication.class, args);
20
21         LOGGGER.info("Application Started Successfully");
22     }
23 }
```

1. src/main/java

This folder contains the main Java source code of the application, including the SpringLearnApplication.java class, which is the entry point of the project.

2. src/main/resources

This folder stores application configuration files such as application.properties and other resources used by the application.

3. src/test/java

This folder contains test classes used to verify that the application works correctly. Spring Boot creates a default test class for checking if the application context loads successfully.

4. SpringLearnApplication.java – main() Method

The main() method is the starting point of the Spring Boot application. It calls SpringApplication.run() to initialize the Spring container and start the embedded Tomcat server.

5. Purpose of @SpringBootApplication

@SpringBootApplication is the main annotation used in Spring Boot. It combines:

- @Configuration – Defines configuration.
- @EnableAutoConfiguration – Automatically configures the application.
- @ComponentScan – Scans for Spring components.

6. pom.xml

Project Configuration

- groupId: Identifies the organization (com.cognizant).
- artifactId: Specifies the project name (spring-learn).
- version: Defines the project version.

Dependencies

- Spring Web: Provides web and REST API support with an embedded Tomcat server.
- Spring Boot DevTools: Enables automatic restart during development.
- Spring Boot Test: Provides libraries for unit and integration testing.

Build

The build section contains the Spring Boot Maven Plugin used to package the application as an executable JAR.

7. Dependency Hierarchy

The dependency hierarchy displays all the libraries used by the project and their dependent libraries. It can be viewed using:

```
mvn dependency:tree
```

Spring Core – Load Country from Spring Configuration XML

```

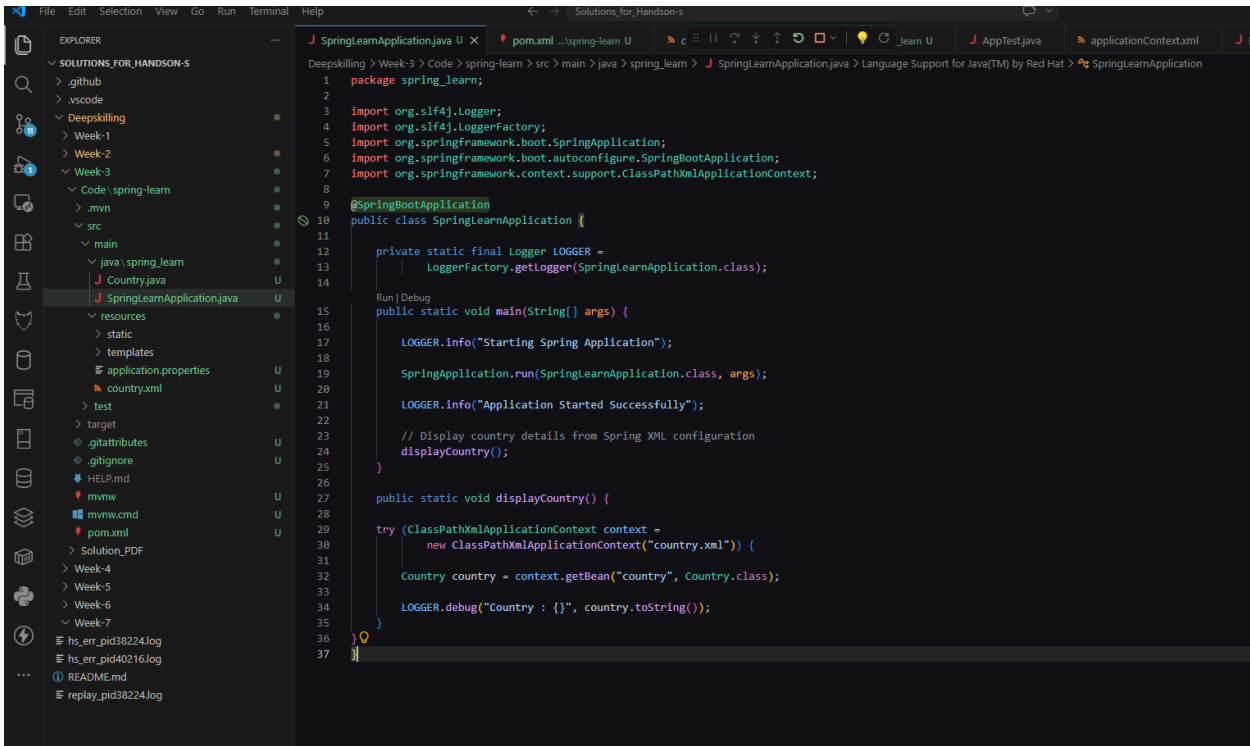
PS D:\Cogniz_folder\Solutions_for_Handson> cd "D:\Cogniz_folder\Solutions_for_Handson-s" & 'C:\Program Files\Java\jdk-17.0.18\bin\java.exe' '%C:\Users\uckal\AppData\Local\Temp\cp_Buaj48ovok81Syfhu1v2a78t.argfile' 'spring_learn.SpringLearnApplication'
00:18:36.184 [main] INFO spring_learn.SpringLearnApplication -- Starting Spring Application
00:18:36.417 [restartMain] INFO spring_learn.SpringLearnApplication -- Starting Spring Application

:: Spring Boot :: (v4.1.1-SNAPSHOT)

2026-07-05T00:18:36.910+05:30 INFO 5048 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.18 with PID 5048 (D:\Cogniz_folder\Solutions_for_Handson-s\DeepskillingWeek-3\Code\Spring-learn\target\classes started by uckal in D:\Cogniz_folder\Solutions_for_Handson-s)
2026-07-05T00:18:36.913+05:30 INFO 5048 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T00:18:37.046+05:30 INFO 5048 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-05T00:18:37.047+05:30 INFO 5048 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-05T00:18:38.236+05:30 INFO 5048 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-05T00:18:38.251+05:30 INFO 5048 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T00:18:38.252+05:30 INFO 5048 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T00:18:38.292+05:30 INFO 5048 --- [spring-learn] [ restartedMain] b.w.c.s.web.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1243 ms
2026-07-05T00:18:39.311+05:30 INFO 5048 --- [spring-learn] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 1 endpoint beneath base path '/actuator'
2026-07-05T00:18:39.408+05:30 INFO 5048 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-05T00:18:39.410+05:30 INFO 5048 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.994 seconds (process running for 3.655)
2026-07-05T00:18:39.426+05:30 INFO 5048 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Application Started Successfully
2026-07-05T00:18:39.536+05:30 WARN 5048 --- [spring-learn] [ restartedMain] o.s.c.s.ClassPathXmlApplicationContext : Exception encountered during context initialization - cancelling refresh attempt
: org.springframework.beans.factory.CannotLoadBeanClassException: Cannot find class [com.cognizant.springlearn.Country] for bean with name 'country' defined in class path resource [country.xml]
Exception in thread "restartMain" java.lang.reflect.InvocationTargetException
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:77)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:569)
    at org.springframework.boot.devtools.restart.RestartLauncher.run(RestartLauncher.java:52)
Caused by: org.springframework.beans.factory.CannotLoadBeanClassException: Cannot find class [com.cognizant.springlearn.Country] for bean with name 'country' defined in class path resource [country.xml]
    at org.springframework.beans.factory.support.AbstractBeanFactory.resolveBeanClass(AbstractBeanFactory.java:1571)
    at org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.determineTargetType(AbstractAutowireCapableBeanFactory.java:684)
    at org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.predictBeanType(AbstractAutowireCapableBeanFactory.java:653)
    at org.springframework.beans.factory.support.AbstractBeanFactory.isFactoryBean(AbstractBeanFactory.java:1699)
    at org.springframework.beans.factory.support.AbstractBeanFactory.isFactoryBean(AbstractBeanFactory.java:1162)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.instantiateSingleton(DefaultListableBeanFactory.java:1218)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.preInstantiateSingleton(DefaultListableBeanFactory.java:1191)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.preInstantiateSingletons(DefaultListableBeanFactory.java:1121)
    at org.springframework.context.support.AbstractApplicationContext.finishBeanFactoryInitialization(AbstractApplicationContext.java:994)
    at org.springframework.context.support.AbstractApplicationContext.refresh(AbstractApplicationContext.java:621)
    at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:144)
    at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:85)

Caused by: org.springframework.beans.factory.CannotLoadBeanClassException: Cannot find class [com.cognizant.springlearn.Country] for bean with name 'country' defined in class path resource [country.xml]
    at org.springframework.beans.factory.support.AbstractBeanFactory.resolveBeanClass(AbstractBeanFactory.java:1571)
    at org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.determineTargetType(AbstractAutowireCapableBeanFactory.java:684)
    at org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.predictBeanType(AbstractAutowireCapableBeanFactory.java:653)
    at org.springframework.beans.factory.support.AbstractBeanFactory.isFactoryBean(AbstractBeanFactory.java:1699)
    at org.springframework.beans.factory.support.AbstractBeanFactory.isFactoryBean(AbstractBeanFactory.java:1162)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.instantiateSingleton(DefaultListableBeanFactory.java:1218)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.preInstantiateSingleton(DefaultListableBeanFactory.java:1191)
    at org.springframework.beans.factory.support.DefaultListableBeanFactory.preInstantiateSingletons(DefaultListableBeanFactory.java:1121)
    at org.springframework.context.support.AbstractApplicationContext.finishBeanFactoryInitialization(AbstractApplicationContext.java:994)
    at org.springframework.context.support.AbstractApplicationContext.refresh(AbstractApplicationContext.java:621)
    at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:144)
    at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:85)
    at spring_learn.SpringLearnApplication.displayCountry(SpringLearnApplication.java:38)
    at spring_learn.SpringLearnApplication.main(SpringLearnApplication.java:24)
    ... 5 more
Caused by: java.lang.ClassNotFoundException: com.cognizant.springlearn.Country
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
    at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
    at java.base/java.lang.Class.forName0(Native Method)
    at java.base/java.lang.Class.forName(Class.java:469)
    at org.springframework.boot.devtools.restart.RestartClassLoader.loadClass(RestartClassLoader.java:123)
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
    at java.base/java.lang.Class.forName0(Native Method)
    at java.base/java.lang.Class.forName(Class.java:469)
    at org.springframework.util.ClassUtils.forName(ClassUtils.java:388)
    at org.springframework.beans.factory.support.AbstractBeanDefinition.resolveBeanClass(AbstractBeanDefinition.java:489)
    at org.springframework.beans.factory.support.AbstractBeanFactory.doResolveBeanClass(AbstractBeanFactory.java:1638)
    at org.springframework.beans.factory.support.AbstractBeanFactory.resolveBeanClass(AbstractBeanFactory.java:1564)
    ... 18 more
2026-07-05T00:18:39.756+05:30 INFO 5048 --- [spring-learn] [on(2)-127.0.0.1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T00:18:39.757+05:30 INFO 5048 --- [spring-learn] [on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T00:18:39.769+05:30 INFO 5048 --- [spring-learn] [on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms

```



1. Bean Tag

The <bean> tag is used to define an object that is managed by the Spring Framework. When the application starts, Spring reads the XML configuration file, creates the bean, and manages its lifecycle.

Example:

```
<bean id="country" class="spring_learn.Country">
```

2. id Attribute

The id attribute gives a unique name to the bean. This name is used to retrieve the bean from the Spring container using the getBean() method.

Example:

```
id="country"
```

3. class Attribute

The class attribute specifies the fully qualified name of the Java class whose object will be created by Spring.

Example:

```
class="spring_learn.Country"
```

When the application starts, Spring creates an object of the Country class.

4. Property Tag

The <property> tag is used to assign values to the bean's properties. Spring injects these values by calling the corresponding setter methods.

Example:

```
<property name="code" value="IN"/>
<property name="name" value="India"/>
```

5. name Attribute

The name attribute specifies the property of the bean that should receive the value. It corresponds to the variable name in the Java class.

Example:

```
name="code"
```

This causes Spring to call:

```
setCode("IN");
```

6. value Attribute

The value attribute specifies the actual value that will be assigned to the property.

Example:

```
value="India"
```

This value is passed to the corresponding setter method.

7. ApplicationContext

ApplicationContext is the central interface of the Spring Framework. It acts as the Spring IoC (Inversion of Control) container, which is responsible for creating, configuring, and managing application beans.

It loads the configuration, creates objects, injects dependencies, and makes the beans available to the application.

8. ClassPathXmlApplicationContext

ClassPathXmlApplicationContext is an implementation of the ApplicationContext interface. It loads the Spring XML configuration file from the application's classpath.

Example:

```
ApplicationContext context =
    new ClassPathXmlApplicationContext("country.xml");
```

In this program, it reads the country.xml file and creates the beans defined inside it.

9. What Happens When context.getBean() is Invoked?

When the following statement is executed:

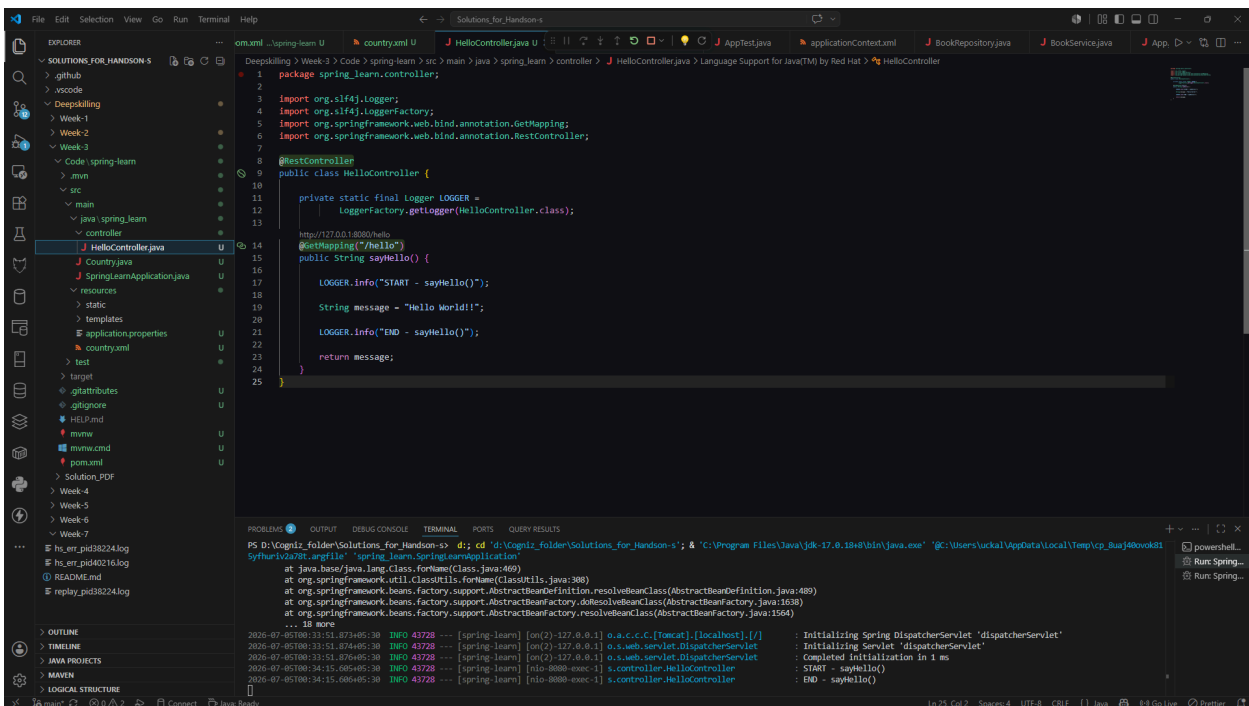
```
Country country = context.getBean("country", Country.class);
```

Spring performs the following steps:

1. It searches the XML configuration file for the bean with the ID country.
2. It creates an object of the Country class if one has not already been created.
3. It invokes the default constructor of the Country class.
4. It calls the setter methods (setCode() and setName()) to inject the values specified in the XML file.
5. It returns the fully initialized Country object to the application.

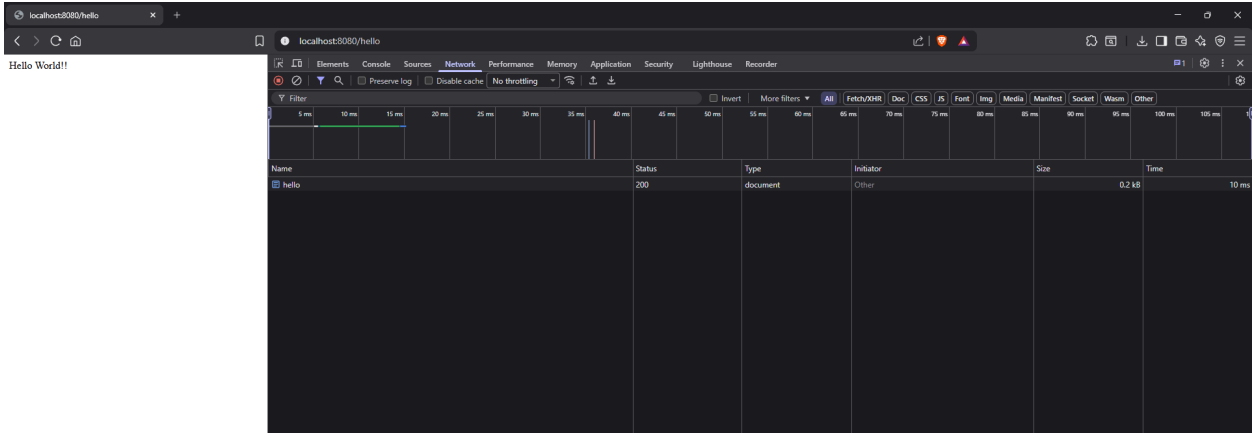
2. Spring-rest-handson

Hello World RESTful Web Service



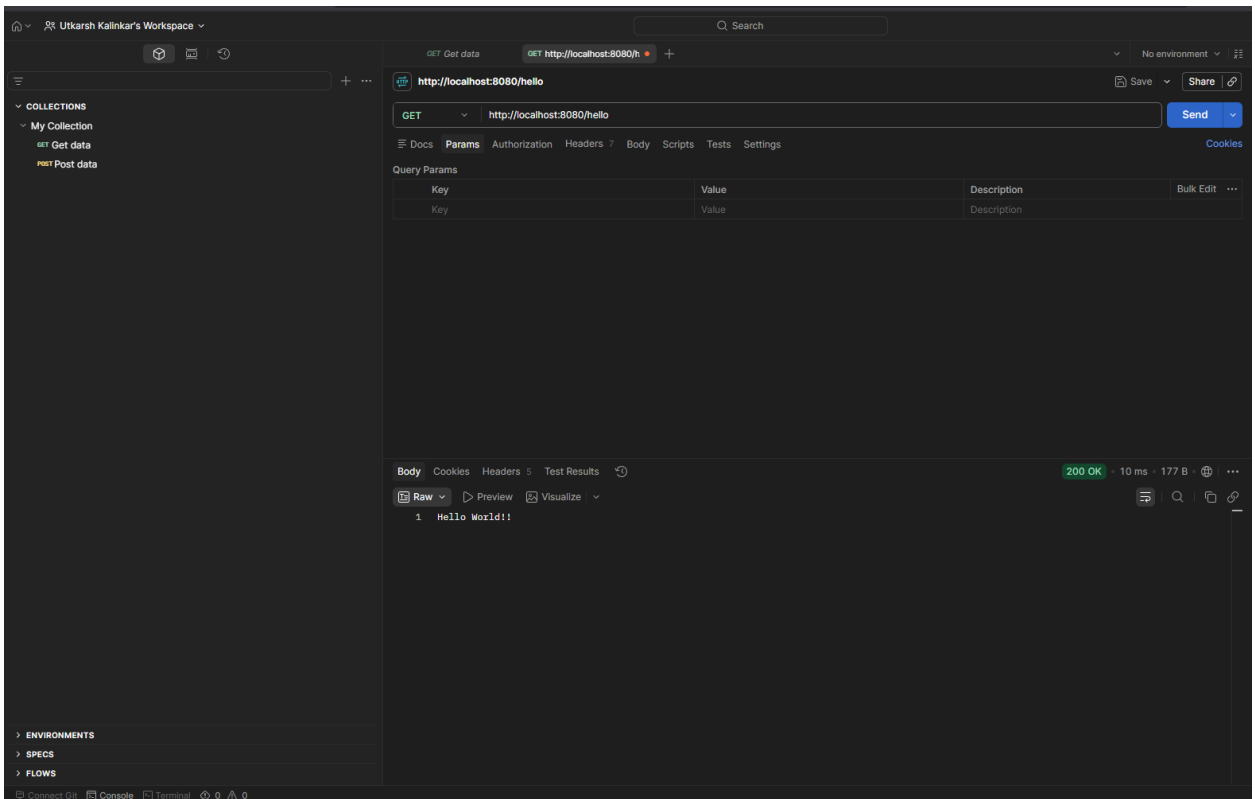
HTTP Header Details (Browser)

The Network tab displays the HTTP request and response headers exchanged between the browser and the server.

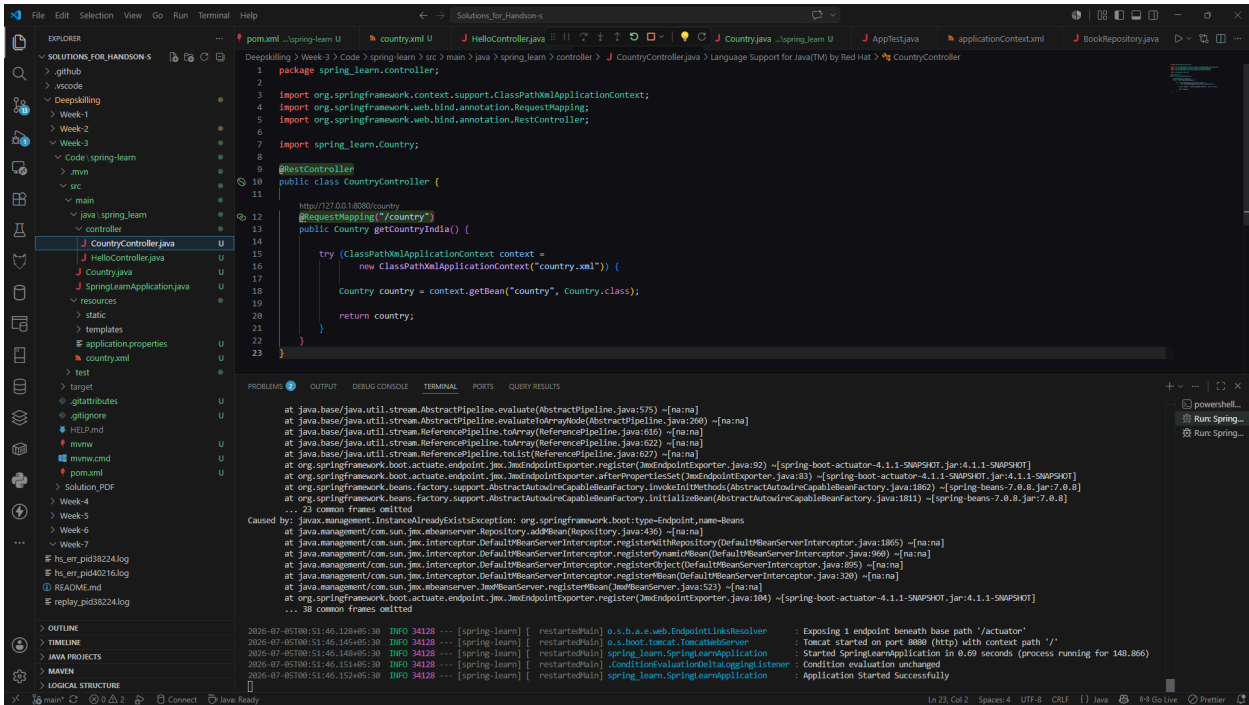


HTTP Header Details (Postman)

The Headers tab in Postman displays the response headers returned by the Spring Boot application.



REST - Country Web Service



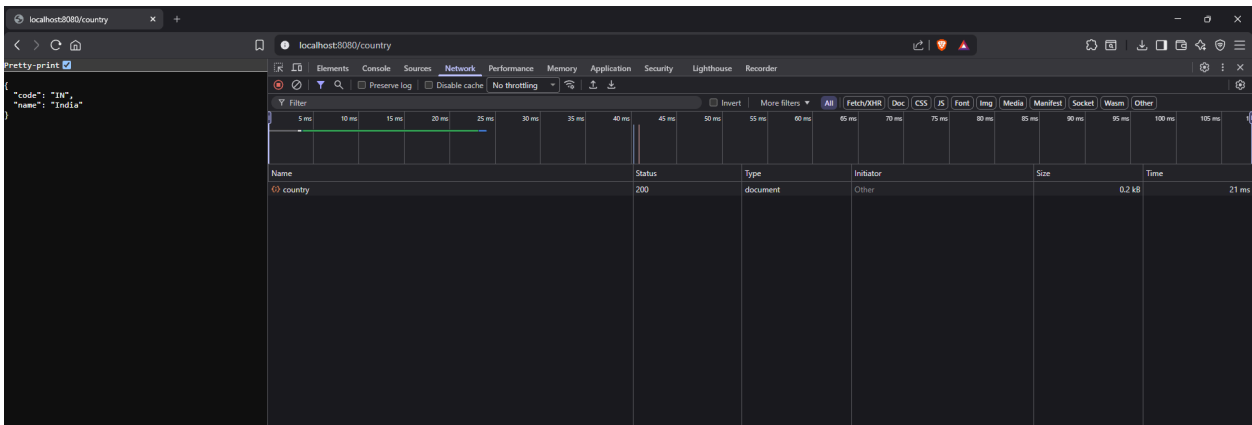
1. What happens in the controller method?

When a client sends a request to `/country`, Spring calls the `getCountryIndia()` method. The method loads the `country.xml` configuration file, retrieves the `Country` bean from the Spring container, and returns the `Country` object to the client.

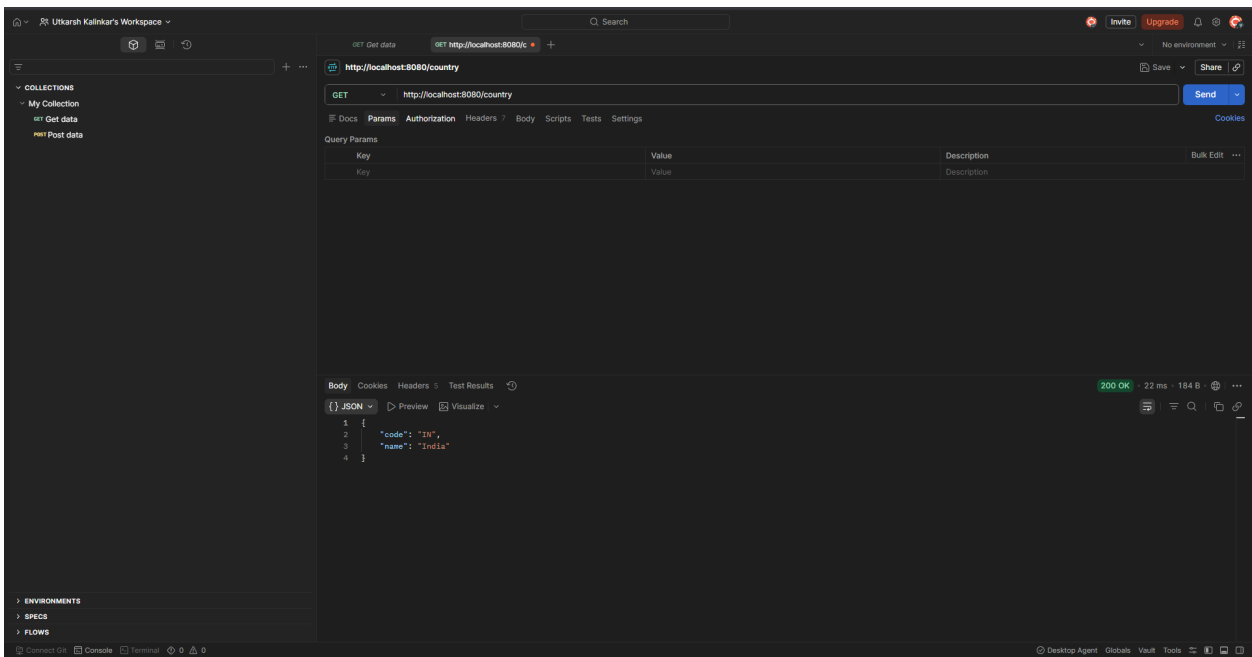
2. How is the bean converted into a JSON response?

The `Country` object returned by the controller is automatically converted into JSON by Spring Boot. This is done by the Jackson library, which is included through the `spring-boot-starter-web` dependency. Jackson reads the object's getter methods and generates the corresponding JSON.

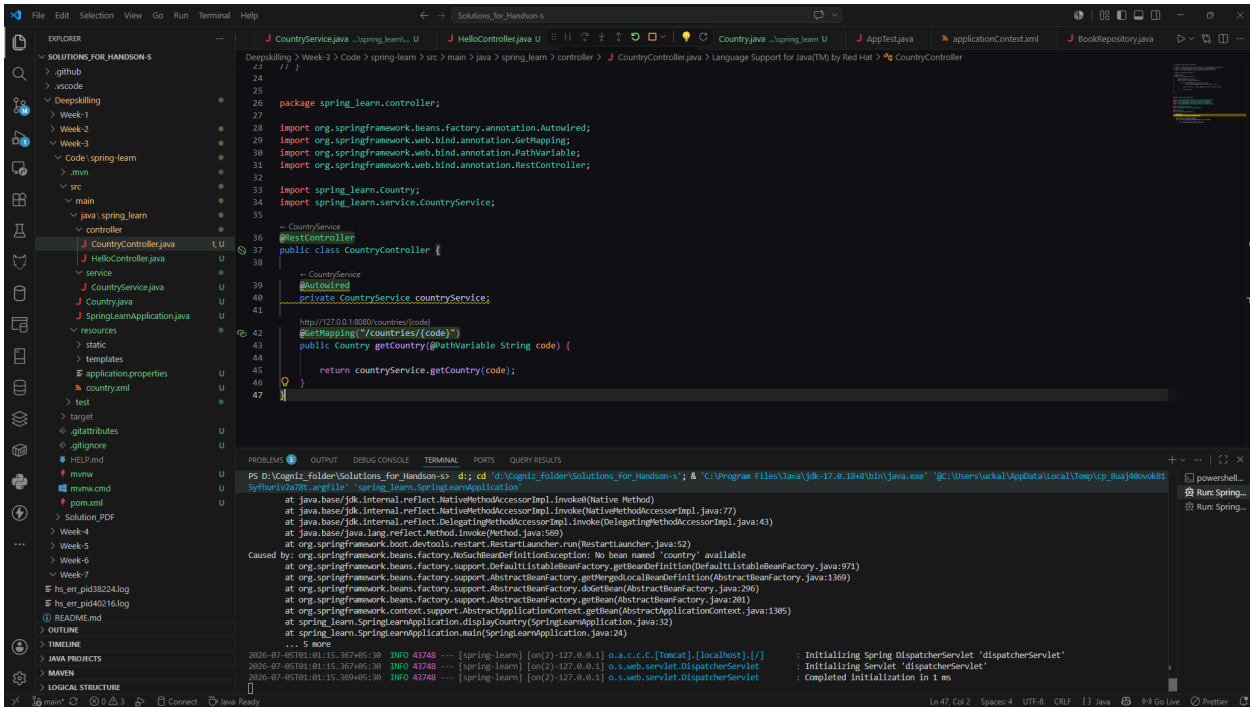
3. Network Tab of Developer Tools



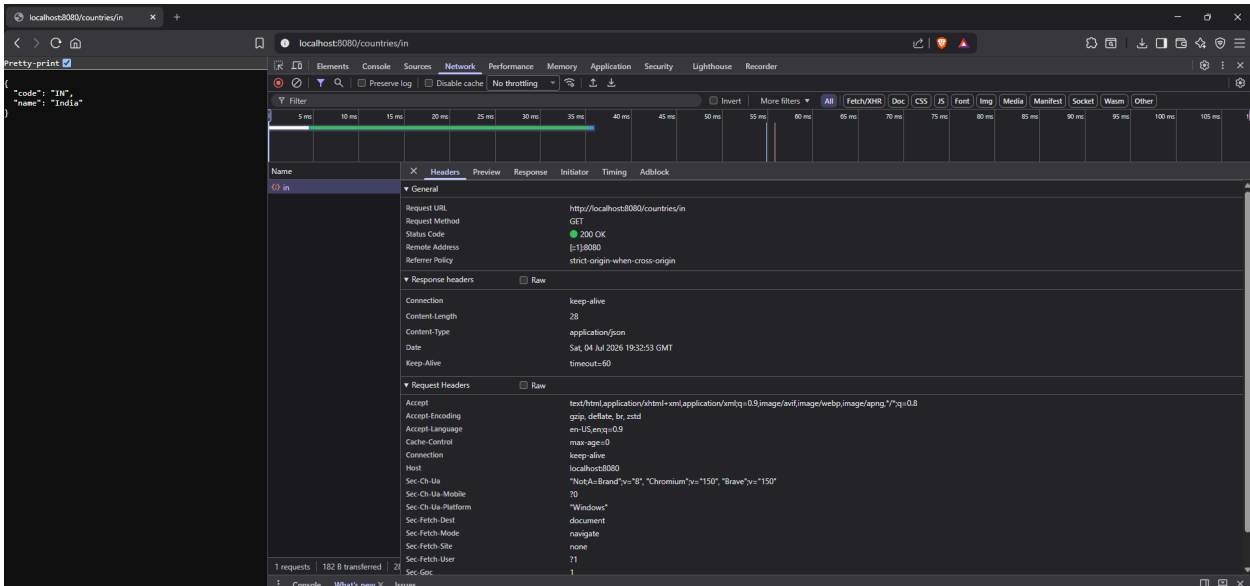
4. Postman Headers



REST - Get country based on country code



Network Tab of Developer Tools



localhost:8080/countries/us

```
pretty-print
{
  "code": "US",
  "name": "United States"
}
```

localhost:8080/countries/us

Filter: 5 ms 10 ms 15 ms 20 ms 25 ms 30 ms 35 ms 40 ms 45 ms 50 ms 55 ms 60 ms 65 ms 70 ms 75 ms 80 ms 85 ms 90 ms 95 ms 100 ms 105 ms

Name: us

General

- Request URL: http://localhost:8080/countries/us
- Request Method: GET
- Status Code: 200 OK
- Remote Address: [::1]:8080
- Referer Policy: strict-origin-when-cross-origin

Response Headers

- Connection: keep-alive
- Content-Length: 36
- Content-Type: application/json
- Date: Sat, 04 Jul 2026 19:34:11 GMT
- Keep-Alive: timeout=60

Request Headers

- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8
- Accept-Encoding: gzip, deflate, br, zstd
- Accept-Language: en-US,en;q=0.9
- Connection: keep-alive
- Host: localhost:8080
- Sec-CH-UA: "NotA=Brand";v="10", "Chromium";v="150", "Brave";v="150"
- Sec-CH-UA-Mobile: 70
- Sec-CH-UA-Platform: "Windows"
- Sec-Fetch-Dest: document
- Sec-Fetch-Mode: navigate
- Sec-Fetch-Site: none
- Sec-Fetch-User: 71
- Sec-Gpc: 1
- Upgrade-Insecure-Requests: 1

1 requests | 190 B transferred | 1 Upgrade-Insecure-Requests

What's new in DevTools 149

localhost:8080/countries/de

```
pretty-print
{
  "code": "DE",
  "name": "Germany"
}
```

localhost:8080/countries/de

Filter: 5 ms 10 ms 15 ms 20 ms 25 ms 30 ms 35 ms 40 ms 45 ms 50 ms 55 ms 60 ms 65 ms 70 ms 75 ms 80 ms 85 ms 90 ms 95 ms 100 ms 105 ms

Name: de

General

- Request URL: http://localhost:8080/countries/de
- Request Method: GET
- Status Code: 200 OK
- Remote Address: [::1]:8080
- Referer Policy: strict-origin-when-cross-origin

Response Headers

- Connection: keep-alive
- Content-Length: 30
- Content-Type: application/json
- Date: Sat, 04 Jul 2026 19:34:33 GMT
- Keep-Alive: timeout=60

Request Headers

- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8
- Accept-Encoding: gzip, deflate, br, zstd
- Accept-Language: en-US,en;q=0.9
- Connection: keep-alive
- Host: localhost:8080
- Sec-CH-UA: "NotA=Brand";v="10", "Chromium";v="150", "Brave";v="150"
- Sec-CH-UA-Mobile: 70
- Sec-CH-UA-Platform: "Windows"
- Sec-Fetch-Dest: document
- Sec-Fetch-Mode: navigate
- Sec-Fetch-Site: none
- Sec-Fetch-User: 71
- Sec-Gpc: 1
- Upgrade-Insecure-Requests: 1

1 requests | 184 B transferred | 1 Upgrade-Insecure-Requests

localhost:8080/countries/jp

```
pretty-print
{
  "code": "JP",
  "name": "Japan"
}
```

localhost:8080/countries/jp

Filter: 5 ms 10 ms 15 ms 20 ms 25 ms 30 ms 35 ms 40 ms 45 ms 50 ms 55 ms 60 ms 65 ms 70 ms 75 ms 80 ms 85 ms 90 ms 95 ms 100 ms 105 ms

Name: jp

General

- Request URL: http://localhost:8080/countries/jp
- Request Method: GET
- Status Code: 200 OK
- Remote Address: [::1]:8080
- Referer Policy: strict-origin-when-cross-origin

Response Headers

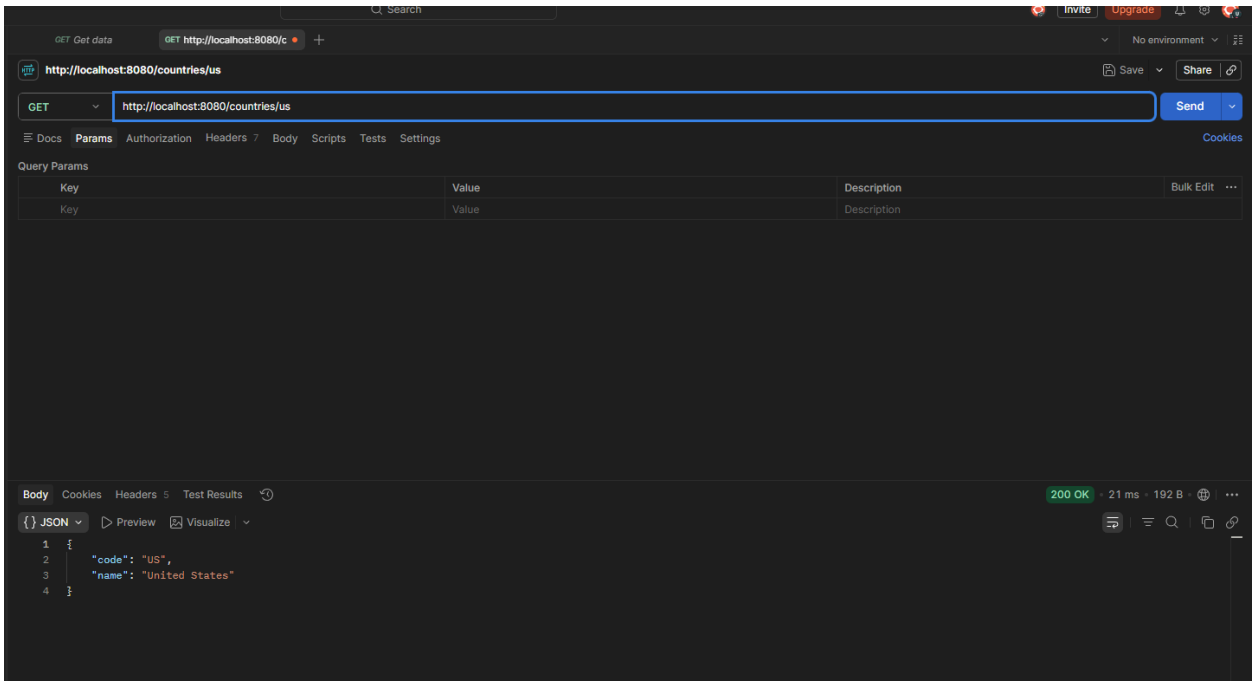
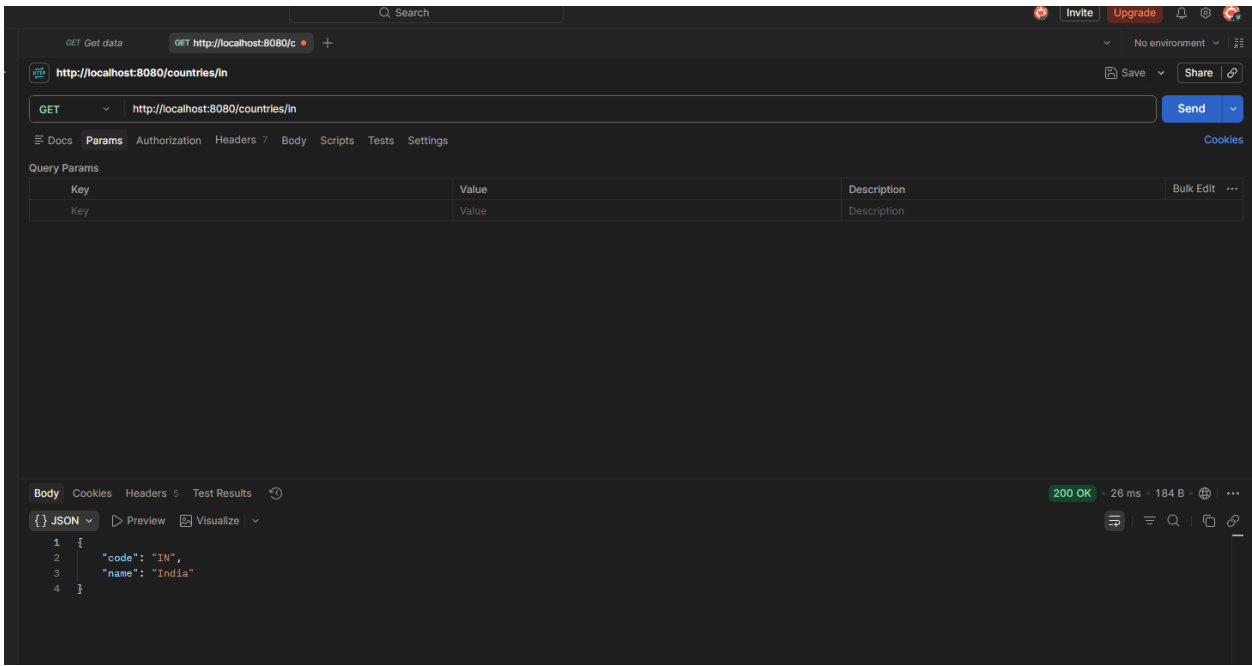
- Connection: keep-alive
- Content-Length: 28
- Content-Type: application/json
- Date: Sat, 04 Jul 2026 19:34:55 GMT
- Keep-Alive: timeout=60

Request Headers

- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8
- Accept-Encoding: gzip, deflate, br, zstd
- Accept-Language: en-US,en;q=0.9
- Connection: keep-alive
- Host: localhost:8080
- Sec-CH-UA: "NotA=Brand";v="10", "Chromium";v="150", "Brave";v="150"
- Sec-CH-UA-Mobile: 70
- Sec-CH-UA-Platform: "Windows"
- Sec-Fetch-Dest: document
- Sec-Fetch-Mode: navigate
- Sec-Fetch-Site: none
- Sec-Fetch-User: 71
- Sec-Gpc: 1
- Upgrade-Insecure-Requests: 1

1 requests | 182 B transferred | 1 Upgrade-Insecure-Requests

Postman Headers



GET Get data +

http://localhost:8080/countries/de Save Share

GET Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 5 Test Results 200 OK · 22 ms · 186 B

{ } JSON Preview Visualize

```
1 {
2   "code": "DE",
3   "name": "Germany"
4 }
```

GET Get data +

http://localhost:8080/countries/jp Save Share

GET Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 5 Test Results 200 OK · 22 ms · 184 B

{ } JSON Preview Visualize

```
1 {
2   "code": "JP",
3   "name": "Japan"
4 }
```

5. JWT-handson

Create authentication service that returns JWT

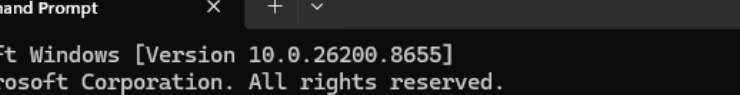
- **Create authentication controller and configure it in SecurityConfig**

```

: Spring Boot : (v4.1.1-SNAPSHOT)

2026-07-05T01:28:54.896405:30 INFO I1692 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.18 with PID 11692 (D:\Cogniz-fol
ider\Solutions-for_Handson->DeepDillingWeek-3\Code\SpringLearn\target\classes started by ucak in D:\Cogniz-fol
2026-07-05T01:28:54.896405:30 DEBUK I1692 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Running with Spring Boot v4.1.1-SNAPSHOT, Spring v7.0.8
2026-07-05T01:28:54.897405:30 INFO I1692 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T01:28:54.353405:30 INFO I1692 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
2026-07-05T01:28:54.354405:30 INFO I1692 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T01:28:54.354405:30 INFO I1692 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T01:28:54.377405:30 INFO I1692 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: Initialization completed in 279 ms
2026-07-05T01:28:54.428405:30 INFO I1692 --- [spring-learn] [ restartedMain] r.InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsServiceImpl bean with name
userDetailsService
2026-07-05T01:28:54.527405:30 INFO I1692 --- [spring-learn] [ restartedMain] o.s.b.a.w.eb.EndpointLinksResolver : Exposing 1 endpoint beneath base path '/actuator'
2026-07-05T01:28:54.543405:30 INFO I1692 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
2026-07-05T01:28:54.545405:30 INFO I1692 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Started SpringLearnApplication in 0.478 seconds (process running for 384.369)
2026-07-05T01:28:54.549405:30 INFO I1692 --- [spring-learn] [ restartedMain] .ConditionEvaluator.onDeltaLoggingListener : Condition evaluation unchanged
2026-07-05T01:28:54.549405:30 INFO I1692 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Application Started Successfully
2026-07-05T01:29:34.852405:30 INFO I1692 --- [spring-learn] [nio-8090-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T01:29:34.852405:30 INFO I1692 --- [spring-learn] [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T01:29:34.834405:30 INFO I1692 --- [spring-learn] [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-05T01:29:35.006405:30 INFO I1692 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : START
2026-07-05T01:29:35.006405:30 DEBUK I1692 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Authorization Header : Basic dG91c2QudjZQ==
2026-07-05T01:29:35.008405:30 INFO I1692 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : END

```



The screenshot shows a Windows Command Prompt window with the title bar 'Command Prompt'. The window content displays the following text:

```
Microsoft Windows [Version 10.0.26200.8655]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\uckal>curl -s -u user:pwd http://localhost:8090/authenticate  
{ "token": "" }  
C:\Users\uckal>
```

```

1 package spring_learn.controller;
2
3 import java.util.HashMap;
4 import java.util.Map;
5
6 import org.slf4j.Logger;
7 import org.slf4j.LoggerFactory;
8 import org.springframework.web.bind.annotation.GetMapping;
9 import org.springframework.web.bind.annotation.RequestHeader;
10 import org.springframework.web.bind.annotation.RestController;
11
12 @RestController
13 public class AuthenticationController {
14
15     private static final Logger LOGGER =
16         LoggerFactory.getLogger(AuthenticationController.class);
17
18     @GetMapping("/authenticate")
19     public Map<String, String> authenticate(
20         @RequestHeader("Authorization") String authHeader) {
21
22         LOGGER.info("START");
23
24         LOGGER.debug("Authorization Header : {}", authHeader);
25
26         Map<String, String> map = new HashMap<>();
27
28         map.put(key: "token", value: "");
29
30         LOGGER.info("END");
31
32         return map;
33     }
34 }

```

• Read Authorization Header and Decode the Username

```
PS D:\Cogniz_folder\Solutions_for_Hanson-s> d; cd 'd:\Cogniz_folder\Solutions_for_Hanson-s'; & "C:\Program Files\Java\jdk-17.0.18\bin\java.exe" "@C:\Users\uckal\AppData\Local\Temp\cp_md327\okdes
0159g21rj3o.urf1le" "spring_learn.SpringLearnApplication"

:: Spring Boot :: (v4.1.1-SNAPSHOT)

2026-07-05T01:40:37.996405:30 INFO 42356 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.18 with PID 42356 (D:\Cogniz_fo
Iden\Solutions_for_Hanson-s\Deeplearning\Week-3\Code\spring_learn\target\classes started by uckal in D:\Cogniz_folder\Solutions_for_Hanson-s)
2026-07-05T01:40:37.996405:30 DEBUG 42356 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.1-SNAPSHOT, Spring v7.0.8
2026-07-05T01:40:38.238405:30 INFO 42356 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T01:40:38.239405:30 INFO 42356 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
2026-07-05T01:40:38.239405:30 INFO 42356 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T01:40:38.240405:30 INFO 42356 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T01:40:38.271405:30 INFO 42356 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: Initialization completed in 273 ms
2026-07-05T01:40:38.351405:30 INFO 42356 --- [spring-learn] [ restartedMain] r.InitializedUserDetailsServiceManagerConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name
userDetailsService
2026-07-05T01:40:38.474405:30 INFO 42356 --- [spring-learn] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 1 endpoint beneath base path '/actuator'
2026-07-05T01:40:38.485405:30 INFO 42356 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
2026-07-05T01:40:38.489405:30 INFO 42356 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.511 seconds (process running for 93.165)
2026-07-05T01:40:38.490405:30 INFO 42356 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition evaluation unchanged
2026-07-05T01:40:38.491405:30 INFO 42356 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Application Started Successfully
2026-07-05T01:41:00.167405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T01:41:00.167405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T01:41:00.168405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-05T01:41:00.302405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : START
2026-07-05T01:41:00.302405:30 DEBUG 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Authorization Header : Basic dXNlcjpwd2Q=
2026-07-05T01:41:00.302405:30 DEBUG 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Inside getUser()
2026-07-05T01:41:00.303405:30 DEBUG 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Encoded Credentials : dXNlcjpwd2Q=
2026-07-05T01:41:00.303405:30 DEBUG 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Decoded Credentials : user:pwd
2026-07-05T01:41:00.304405:30 DEBUG 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : User : user
2026-07-05T01:41:00.304405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : Authenticated User : user
2026-07-05T01:41:00.304405:30 INFO 42356 --- [spring-learn] [nio-8090-exec-1] s.controller.AuthenticationController : END
[]
```

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

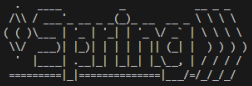
C:\Users\uckal>curl -s -u user:pwd http://localhost:8090/authenticate
{"token":""}
C:\Users\uckal>
```

```
Deeplearning > Week-3 > Code > spring_learn > src > main > java > spring_learn > controller > AuthenticationController.java > Language Support for Java(TM) by Red Hat > AuthenticationController > getUser(String)

16 public class AuthenticationController {
17     private static final Logger LOGGER =
18         LoggerFactory.getLogger(AuthenticationController.class);
19
20     http://127.0.0.1:8090/authenticate
21     @GetMapping("/authenticate")
22     public Map<String, String> authenticate(
23         @RequestHeader("Authorization") String authHeader) {
24
25         LOGGER.info("START");
26
27         LOGGER.debug("Authorization Header : {}", authHeader);
28
29         String user = getUser(authHeader);
30
31         LOGGER.debug("Authenticated User : {}", user);
32
33         Map<String, String> map = new HashMap<>();
34
35         map.put(key: "token", value: "");
36
37         LOGGER.info("END");
38
39         return map;
40     }
41
42     private String getUser(String authHeader) {
43
44         LOGGER.debug("Inside getUser()");
45
46         String encodedCredentials = authHeader.substring(beginIndex: 0);
47
48         LOGGER.debug("Encoded Credentials : {}", encodedCredentials);
49
50         byte[] decodedBytes = Base64.getDecoder().decode(encodedCredentials);
51
52         String credentials = new String(decodedBytes, StandardCharsets.UTF_8);
53
54         LOGGER.debug("Decoded Credentials : {}", credentials);
55
56         String user = credentials.substring(beginIndex: 0, credentials.indexOf(":"));
57
58         LOGGER.debug("User : {}", user);
59
60         return user;
61     }
62 }
```

• Generate token based on the user

```
PS D:\Cogniz_folder\Solutions_for_Handson> d; cd 'd:\Cogniz_folder\Solutions_for_Handson-s'; & 'C:\Program Files\Java\jdk-17.0.18\bin\java.exe' '@C:\Users\uckal\AppData\Local\Temp\cp_9xe6rvtjdj143h8aizjpv5zo-argfile' 'spring_learn.SpringLearnApplication'
```



```

:: Spring Boot ::      (v4.1.1-SNAPSHOT)

2026-07-05T01:50:38.694+05:30 INFO 3732 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.18 with PID 3732 (D:\Cogniz_fold
er\Solutions_for_Handson-s\DeepsKillingWeek-3\Code\spring_learn\target\classes started by uckal in D:\Cogniz_folder\Solutions_for_Handson-s)
2026-07-05T01:50:38.699+05:30 DEBUG 3732 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Running with Spring Boot v4.1.1-SNAPSHOT, Spring v7.0.8
2026-07-05T01:50:38.708+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.s.b.devtools.restart.ChangedReloader : No active profile set, falling back to 1 default profile: "default"
2026-07-05T01:50:38.769+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.s.b.devtools.restart.ChangedReloader : The Class-Path manifest attribute in C:\Users\uckal\.m2\repository\org\glassfish
\jaxb\jaxb-runtime\2.3.1\jaxb-runtime-2.3.1.jar referenced one or more files that do not exist: file:/C:/Users/uckal/.m2/repository/org/glassfish/jaxb/jaxb-runtime/2.3.1/jaxb-api-2.3.1.jar, file:/C:/Use
rs/uckal/.m2/repository/org/glassfish/jaxb/jaxb-runtime/2.3.1/stax-ex-1.8.jar, file:/C:/Users/uckal/.m2/repository/org/glassfish/jaxb/jaxb-runtime/2.3.1/istack-commons-runtime-3.0.7.jar, file:/C:/Users/uc
kal/.m2/repository/org/glassfish/jaxb/jaxb-runtime/2.3.1/javax.activation-api-1.2.0.jar
2026-07-05T01:50:38.761+05:30 INFO 3732 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'fals
e' to disable
2026-07-05T01:50:38.762+05:30 INFO 3732 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' prop
erty to 'DEBUG'
2026-07-05T01:50:40.085+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
2026-07-05T01:50:40.099+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T01:50:40.099+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T01:50:40.148+05:30 INFO 3732 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: Initialization completed in 1386 ms
2026-07-05T01:50:40.589+05:30 INFO 3732 --- [spring-learn] [ restartedMain] r$Initializer$UserDetailsServiceManagerConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name u
serDetailsService
2026-07-05T01:50:41.618+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.s.b.a.e.web.EndpointLinksResolver : Exposing 1 endpoint beneath base path '/actuator'
2026-07-05T01:50:41.755+05:30 INFO 3732 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
2026-07-05T01:50:41.779+05:30 INFO 3732 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 3.505 seconds (process running for 4.255)
2026-07-05T01:50:41.801+05:30 INFO 3732 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Application Started Successfully
2026-07-05T01:50:44.288+05:30 INFO 3732 --- [spring-learn] [ on(2)-127.0.0.1] o.a.c.c.c.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T01:50:44.288+05:30 INFO 3732 --- [spring-learn] [ on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T01:50:44.289+05:30 INFO 3732 --- [spring-learn] [ on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
2026-07-05T01:51:05.024+05:30 INFO 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : START
2026-07-05T01:51:05.025+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Authorization Header : Basic dXNlcjpwd2Q=
2026-07-05T01:51:05.025+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Inside getUser()
2026-07-05T01:51:05.025+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Encoded Credentials : dXNlcjpwd2Q=
2026-07-05T01:51:05.026+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Decoded Credentials : user:pwd
2026-07-05T01:51:05.026+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : User : user
2026-07-05T01:51:05.026+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Authenticated User : user
2026-07-05T01:51:05.092+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Inside generateJwt()
2026-07-05T01:51:05.092+05:30 DEBUG 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : Generated Token : eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIjIC2YyIiwiaWF0IjoxNzYyZjgqjgqjEBP2zcQDtdat
C1eHAiOiJlE3ODMxOTc2NjY9.MWwY6g2jgqjgqjEBP2zcQDtdatL29W9yYA-PD5rtuf7w0
2026-07-05T01:51:05.093+05:30 INFO 3732 --- [spring-learn] [ nio-8090-exec-1] s.controller.AuthenticationController : END

```

```

Microsoft Windows [Version 10.0.26200.8655]
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C:\Users\uckal>curl -s -u user:pwd http://localhost:8090/authenticate
{"token":"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIjIC2YyIiwiaWF0IjoxNzYyZjgqjgqjEBP2zcQDtdatL29W9yYA-PD5rtuf7w0"}
C:\Users\uckal>

```

```

70
71 private String generateJwt(String user) {
72
73     LOGGER.debug("Inside generateJwt()");
74
75     JwtBuilder builder = Jwts.builder();
76
77     builder.setSubject(user);
78
79     builder.setIssuedAt(new Date());
80
81     builder.setExpiration(new Date(new Date().getTime() + 1200000));
82
83     builder.signWith(SignatureAlgorithm.HS256, "secretkey");
84
85     String token = builder.compact();
86
87     LOGGER.debug("Generated Token : {}", token);
88
89     return token;
90 }
91 }

```